

~\Downloads\Day 24.sql

```
1  -- UDF Task 1 - Employee Bonus Calculation (SQL UDF)
2  CREATE OR REPLACE DATABASE UDF_SEQUENCE_DB;
3
4  USE DATABASE UDF_SEQUENCE_DB;
5  USE SCHEMA PUBLIC;
6
7  CREATE OR REPLACE TABLE EMPLOYEE
8  (
9      EMP_ID NUMBER,
10     EMP_NAME STRING,
11     DEPARTMENT STRING,
12     SALARY NUMBER(10,2)
13 );
14
15
16 INSERT INTO EMPLOYEE VALUES
17 (101, 'John', 'IT', 50000),
18 (102, 'Alice', 'HR', 60000),
19 (103, 'David', 'Finance', 70000),
20 (104, 'Neha', 'Sales', 45000),
21 (105, 'Ravi', 'IT', 55000),
22 (106, 'Priya', 'HR', 65000),
23 (107, 'Akash', 'Finance', 75000),
24 (108, 'Sneha', 'Sales', 52000);
25
26 CREATE OR REPLACE FUNCTION CALCULATE_BONUS(SAL NUMBER)
27 RETURNS NUMBER(10,2)
28 AS
29 $$
30     SAL * 0.10
31 $$;
32
33 SELECT
34
35 EMP_NAME,
36
37 SALARY,
38
39 CALCULATE_BONUS(SALARY) AS BONUS,
40
41 SALARY + CALCULATE_BONUS(SALARY) AS TOTAL_SALARY
42
43 FROM EMPLOYEE;
44
45
46
47
48 -- UDF Task 2 - Employee Performance Grade
```

```
49 CREATE OR REPLACE TABLE EMPLOYEE_PERFORMANCE
50 (
51     EMP_ID NUMBER,
52     EMP_NAME STRING,
53     RATING NUMBER
54 );
55
56 INSERT INTO EMPLOYEE_PERFORMANCE VALUES
57 (101, 'John', 95),
58 (102, 'Alice', 88),
59 (103, 'David', 76),
60 (104, 'Neha', 65),
61 (105, 'Ravi', 92),
62 (106, 'Priya', 81),
63 (107, 'Akash', 71),
64 (108, 'Sneha', 68),
65 (109, 'Kiran', 84),
66 (110, 'Rohan', 97);
67
68 CREATE OR REPLACE FUNCTION GET_GRADE(RATING NUMBER)
69 RETURNS STRING
70 AS
71 $$
72 CASE
73     WHEN RATING >= 90 THEN 'A'
74     WHEN RATING >= 80 THEN 'B'
75     WHEN RATING >= 70 THEN 'C'
76     ELSE 'D'
77 END
78 $$;
79
80 SELECT
81     EMP_NAME,
82     RATING,
83     GET_GRADE(RATING) AS GRADE
84 FROM EMPLOYEE_PERFORMANCE;
85
86 SELECT
87     GET_GRADE(RATING) AS GRADE,
88     COUNT(*) AS TOTAL_EMPLOYEES
89 FROM EMPLOYEE_PERFORMANCE
90 GROUP BY GET_GRADE(RATING)
91 ORDER BY GRADE;
92
93
94
95
96 -- Sequence Task 1 - Employee Management System
97 CREATE OR REPLACE SEQUENCE EMP_SEQ
98 START = 1
```

```
99  INCREMENT = 1;
100
101  CREATE OR REPLACE TABLE EMPLOYEE_AUTO(
102      EMP_ID NUMBER,
103      EMP_NAME STRING,
104      DEPARTMENT STRING
105  );
106
107  INSERT INTO EMPLOYEE_AUTO
108  VALUES
109  (EMP_SEQ.NEXTVAL, 'John', 'IT'),
110  (EMP_SEQ.NEXTVAL, 'Alice', 'HR'),
111  (EMP_SEQ.NEXTVAL, 'David', 'Finance'),
112  (EMP_SEQ.NEXTVAL, 'Neha', 'Sales'),
113  (EMP_SEQ.NEXTVAL, 'Ravi', 'IT'),
114  (EMP_SEQ.NEXTVAL, 'Priya', 'HR'),
115  (EMP_SEQ.NEXTVAL, 'Akash', 'Finance'),
116  (EMP_SEQ.NEXTVAL, 'Sneha', 'Sales'),
117  (EMP_SEQ.NEXTVAL, 'Kiran', 'HR'),
118  (EMP_SEQ.NEXTVAL, 'Rohan', 'IT');
119
120  SELECT *
121  FROM EMPLOYEE_AUTO;
122
123  INSERT INTO EMPLOYEE_AUTO
124  VALUES
125  (EMP_SEQ.NEXTVAL, 'Mohit', 'Sales'),
126  (EMP_SEQ.NEXTVAL, 'Pooja', 'HR'),
127  (EMP_SEQ.NEXTVAL, 'Nikhil', 'IT'),
128  (EMP_SEQ.NEXTVAL, 'Anjali', 'Finance'),
129  (EMP_SEQ.NEXTVAL, 'Varun', 'Sales');
130
131
132  SELECT *
133  FROM EMPLOYEE_AUTO;
134
135
136
137
138  -- Sequence Task 2 - Customer Order Management
139  CREATE OR REPLACE SEQUENCE ORDER_SEQ
140  START = 1000
141  INCREMENT = 10;
142
143  CREATE OR REPLACE TABLE ORDERS
144  (
145      ORDER_ID NUMBER,
146      CUSTOMER_NAME STRING,
147      PRODUCT_NAME STRING,
148      ORDER_DATE DATE,
```

```
149     AMOUNT NUMBER(10,2)
150 );
151
152 INSERT INTO ORDERS
153 VALUES
154 (ORDER_SEQ.NEXTVAL, 'Rahul', 'Laptop', '2026-07-01', 65000),
155 (ORDER_SEQ.NEXTVAL, 'Priya', 'Mouse', '2026-07-01', 800),
156 (ORDER_SEQ.NEXTVAL, 'John', 'Keyboard', '2026-07-02', 1500),
157 (ORDER_SEQ.NEXTVAL, 'Alice', 'Monitor', '2026-07-02', 12000),
158 (ORDER_SEQ.NEXTVAL, 'David', 'Printer', '2026-07-03', 9000),
159 (ORDER_SEQ.NEXTVAL, 'Sneha', 'Laptop', '2026-07-03', 65000),
160 (ORDER_SEQ.NEXTVAL, 'Akash', 'Mouse', '2026-07-04', 800),
161 (ORDER_SEQ.NEXTVAL, 'Neha', 'Keyboard', '2026-07-04', 1500),
162 (ORDER_SEQ.NEXTVAL, 'Ravi', 'Monitor', '2026-07-05', 12000),
163 (ORDER_SEQ.NEXTVAL, 'Kiran', 'Printer', '2026-07-05', 9000),
164 (ORDER_SEQ.NEXTVAL, 'Pooja', 'Laptop', '2026-07-06', 65000),
165 (ORDER_SEQ.NEXTVAL, 'Varun', 'Mouse', '2026-07-06', 800);
166
167 SELECT *
168 FROM ORDERS;
169
170 SELECT MAX(ORDER_ID) AS HIGHEST_ORDER_ID
171 FROM ORDERS;
172
173 SELECT MIN(ORDER_ID) AS LOWEST_ORDER_ID
174 FROM ORDERS;
175
176 SELECT COUNT(*) AS TOTAL_ORDERS
177 FROM ORDERS;
```